

MANISH MAN SHRESTHA

Firmware & Embedded Systems Engineer | IoT Developer

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[Website](#) | [GitHub](#) | [LinkedIn](#)

PROFILE SUMMARY

- Embedded firmware engineer with 15 years of experience delivering real-time, security-hardened firmware across ARM Cortex-M, ESP32, PIC, and 8051 — from board bring-up and RTOS drivers to secure MQTT/TLS telemetry and cloud integration.
- Rare hands-on experience deploying post-quantum cryptography (Kyber512, Dilithium2) and ChaCha20-Poly1305 on resource-constrained ESP32-S3 MCUs — with secure MQTT/TLS telemetry validated over a 120-day field deployment at zero security failures.
- Owens the full stack — silicon to cloud: device drivers, FreeRTOS, Modbus/RS-485 & wireless protocols, PCB bring-up/debug, and C#/.NET + ASP.NET backends for data handling and visualization.
- Led firmware/hardware R&D as a startup CTO; holds a Ph.D. (May 2026) applying secure embedded IoT to real-time agricultural and environmental monitoring.

EDUCATION

Ph.D., Agricultural and Biosystems Engineering <i>South Dakota State University, Brookings, SD, USA</i>	May 2026
Graduate Certificate, Data Science <i>South Dakota State University, Brookings, SD, USA</i>	2025
M.E., Mechatronics Engineering (EQF Level 7) <i>Jeonbuk National University, Jeonju, South Korea</i>	2016
B.E., Electronics and Communication Engineering (EQF Level 6) <i>Cosmos College of Management and Technology, Nepal</i>	2010

CORE FOCUS AREAS

Embedded C/C++ • RTOS & Real-Time Design • Secure IoT & Cryptography • Device Drivers & BSP • Sensor Integration • Wireless Telemetry (MQTT/TLS) • PCB Bring-up & Debug • Firmware Architecture • AI / Machine Learning

TECHNICAL SKILLS

Embedded Firmware: Embedded C/C++, real-time firmware, device drivers, board bring-up, interrupt-driven systems, low-level hardware programming

Languages: C, C++, Python, C#, R

MCUs / SoCs: ARM Cortex-M (STM32, LPC, NXP), ESP32 (ESP-IDF), AVR, PIC, 8051

Platforms: Raspberry Pi (Linux-based development)

RTOS / Real-Time: FreeRTOS, Protothreads, task scheduling, inter-task communication, semaphores, mutexes

Networking: TCP/IP, lwIP, MQTT/MQTTs, Wi-Fi, secure telemetry systems

Protocols: UART, SPI, I2C, 1-Wire, MODBUS RTU

Peripherals: ADC/DAC, PWM, GPIO, Wi-Fi, Bluetooth/BLE, Ethernet, RS-232/485

Embedded Security: TLS, AES, ChaCha20-Poly1305, MQTT over TLS, Post-Quantum Cryptography (Kyber, Dilithium)

Debugging Tools: JTAG/SWD, oscilloscopes, logic analyzers, serial debugging, firmware trace analysis, Git

Dev Platforms: ESP-IDF, Keil MDK, MPLAB, Visual Studio, Altium Designer, Eagle, Azure IoT Hub

Memory / Storage: EEPROM, NOR/NAND Flash, SPI Flash, NVS, bootloaders, OTA-safe firmware updates

Low-level Systems: Interrupts, DMA, memory-mapped I/O, ISR design, real-time scheduling

Signal Processing: MATLAB (DSP, filter design, FFT, IIR/FIR), embedded DSP integration

PROFESSIONAL EXPERIENCE

Graduate Research Assistant <i>South Dakota State University, Brookings, SD, USA</i>	Jan 2023 – Present
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Key Responsibilities

- Conducted research on secure and post-quantum IoT-based monitoring systems for real-time agricultural sensing, including soil moisture, environmental parameters, and nutrient (nitrate) dynamics.
- Designed integrated sensing, communication, and data-processing architectures for precision irrigation and nutrient management in controlled-environment agriculture.

Significant Accomplishments

- Designed and deployed embedded IoT monitoring systems for greenhouse and hydroponic environments, enabling continuous real-time measurement of environmental, soil moisture, and nutrient parameters.
- Developed custom multi-sensor edge nodes using ESP32-S3, integrating environmental sensors, soil moisture sensors, and electrochemical nutrient sensing systems.
- Conducted controlled-environment agriculture experiments involving nitrate, chloride, soil moisture, environmental monitoring, and cyanobacteria-based moisture retention, contributing to three manuscripts currently under review.
- Built a secure telemetry pipeline using AES, TLS (MQTTS), and reliability mechanisms (e.g., exponential backoff) for robust and fault-tolerant data transmission.
- Implemented post-quantum cryptography (Kyber512, Dilithium2) on a 240 MHz ESP32-S3, benchmarking ~8.3 ms Kyber key encapsulation and ~84 ms Dilithium signing — demonstrating PQC is viable on low-cost, resource-constrained MCUs.
- Integrated ChaCha20-Poly1305 authenticated encryption (~0.3 ms per packet) for low-overhead payload protection within the communication stack.
- Designed a distributed IoT architecture with 3 synchronized ESP32-S3 sensing nodes for moisture, nitrate, and environmental monitoring.
- Developed backend systems for secure data handling: a C# application for telemetry retrieval, cryptographic verification, and decryption, and an ASP.NET dashboard for real-time visualization and analysis.
- Validated the post-quantum-secured system over a 120-day hydroponic field deployment with 0 signature or decryption failures (~9.4 kB packets, ~8.4 MB free heap), proving production reliability without compromising sensing accuracy.
- Developed a general-purpose secure IoT datalogger architecture supporting Modbus (RS-485) and ADC-based sensors, applicable to industrial and environmental monitoring systems.
- Completed Ph.D. research with one peer-reviewed publication and three additional manuscripts under review on secure IoT systems, post-quantum-secured sensing architectures, and real-time agricultural monitoring.

Teaching Assistant

Jan 2023 – Dec 2025

South Dakota State University, Brookings, SD, USA

Key Responsibilities

- Supported the Engineering Properties of Biological Materials Lab — set up moisture- and temperature-sensor measurements, used oscilloscopes for signal verification, and guided students through data acquisition and analysis.

Assistant Professor

Aug 2017 – Dec 2022

Cosmos College of Management and Technology, Lalitpur, Nepal

Key Responsibilities

- Taught undergraduate courses in Industrial Electronics and Drives, Digital Communication, Control Systems, and Embedded Systems.
- Played a key role in establishing and advancing the Cosmos Research Management Cell (Cosmos-RMC).
- Co-Principal Investigator — “Development of a Multi-Channel Intelligent Ultrasonic Device for Damage Detection of Metallic Structures,” funded by Pokhara University Research Centre (PURC), April 2018 – January 2020.
- Principal Investigator — “Development of IoT-based Multichannel Low-cost Ultrasonic Device for Health Monitoring of a Metallic Structure,” funded by University Grants Commission, Nepal (UGC), Small Research Development and Innovation Grant 2076-77 (Award No. SRDIG-76/77-Engg-1).
- Principal Investigator — “The Urgency of STEAM Education for Driving KMC as an Innovation Hub,” funded by Kathmandu Metropolitan City – City Planning Commission (Mayor’s Research Fellowship 2020–2021).
- Reviewer and editor of *Advances in Engineering and Technology: An International Journal* (Cosmos College of Management and Technology).

Significant Accomplishments

- Published multiple conference and journal papers in IoT, ultrasonic sensing, and structural health monitoring.
- Designed and developed an intelligent ultrasonic device for damage detection of metallic structures.
- Founding member of the Research Management Cell at Cosmos College of Management and Technology.

CTO / Technical Consultant (Concurrent)

Apr 2017 – Dec 2022

Samadhan Engineering, Lalitpur, Nepal

Key Responsibilities

- Led firmware and hardware R&D focused on STEAM education and hydroponic agriculture systems.
- Implemented real-time motor-control firmware using PWM, timers, ADC feedback, and RTOS scheduling under strict timing constraints.
- Developed embedded control systems involving actuator control, sensor interfacing, communication protocols, and hardware-software integration.
- Designed and reviewed PCB layouts using Altium Designer for embedded sensing and control systems, supporting board bring-up, testing, and hardware-firmware debugging.
- Designed and taught courses in Artificial Neural Networks, IoT, embedded systems, and robotics applications.
- Established Git-based version control and YAML-based CI/CD workflows using Docker images to produce repeatable firmware builds and manage release artifacts.

Significant Accomplishments

- Implemented MQTT, SNTP, and MODBUS-RTU protocols on RTOS.
- Developed a 3-axis servo-controlled robotic arm using hardware-timed PWM generation, I2C-based actuator control, calibrated servo motion, and coordinated multi-joint movement.
- Developed a teleoperated mobile robotic system integrating DC motor control, wireless input handling, live video streaming, and real-time motion-control logic.
- Developed a KIOSK system using Raspberry Pi and Java.

Research Scientist

Mar 2016 – Mar 2017

Measurement Robotics & Diagnostic AI Lab, KAIST, Daejeon, South Korea

Key Responsibilities

- Developed real-time embedded firmware on dual-core ARM Cortex-M4 (LPC4370) for ultrasonic signal generation and 6-channel synchronized data acquisition at 5 MSPS for structural monitoring.
- Implemented FFT, IIR, and FIR filtering algorithms directly on ARM Cortex-M4, enabling real-time embedded signal processing and on-device structural response analysis.
- Built high-throughput data acquisition and wireless communication pipelines using the lwIP TCP/IP stack.

Significant Accomplishments

- Designed and validated an embedded wireless ultrasonic sensing platform for aircraft structural health monitoring.
- Integrated embedded DSP, high-speed data acquisition, and Wi-Fi/TCP-IP communication into a complete real-time monitoring system.
- Published a peer-reviewed journal article based on this work.

Research Assistant

Aug 2013 – Feb 2016

Jeonbuk National University, Jeonju, South Korea

Key Responsibilities

- Developed a laser-based excitation and sensing system for structural diagnostics, integrating actuator control, signal acquisition, and data processing.
- Designed and implemented Qt-based control software to drive laser excitation and coordinate synchronized sensing workflows.
- Prototyped and validated signal-processing algorithms (FFT, IIR, FIR) in MATLAB before integrating them into custom software for near-real-time analysis and visualization.
- Developed embedded firmware for high-speed ultrasonic signal acquisition using PZT sensors, ensuring reliable and synchronized data capture.
- Implemented wireless communication using the lwIP TCP/IP stack for real-time data transmission to host systems.
- Worked across hardware, firmware, and software layers to design and deploy a complete sensing and control system.

Significant Accomplishments

- Built an integrated wireless ultrasonic sensing system combining laser excitation, PZT-based sensing, embedded signal acquisition, and Wi-Fi data transmission.
- Developed control and visualization software for structural diagnostics and ultrasonic response analysis.
- Presented this work at international structural health monitoring and aerospace conferences.

Firmware Engineer

Mar 2011 – Aug 2013

Key Responsibilities

- Developed embedded C/C++ firmware for ARM, PIC, and 8051 microcontrollers, including device drivers, communication stacks, and RTOS-based task scheduling.
- Implemented Protothreads-based cooperative multitasking on 8051 and PIC microcontrollers and developed custom bare-metal schedulers for resource-constrained systems.
- Implemented communication protocols including UART, SPI, Modbus RTU, and cellular telemetry using AT commands.

Significant Accomplishments

- Developed a MODBUS/RTU communication protocol from scratch.
- Developed and delivered firmware for 4 distinct commercial field products — panel meters, MPPT solar battery chargers, queue-management systems, and flood early-warning devices.

ACADEMIC PROJECTS

During Ph.D.

- Monitored soil moisture, temperature, humidity, nitrate, and chloride dynamics using ESP32-S3-based multi-sensor platforms.
- Conducted greenhouse and hydroponic experiments to evaluate irrigation behavior, nutrient dynamics, nitrate leaching, and cyanobacteria-based soil moisture retention.
- Analyzed sensor data in R/Python — statistical modeling and machine learning for data cleaning, feature engineering, and validation — to quantify treatment effects, drydown behavior, nitrate trends, and system performance.
- Evaluated communication reliability, power consumption, firmware efficiency, and cryptographic overhead for secure agricultural IoT deployment.

During M.E.

- Designed and implemented a wireless SHM system using PZT sensors and Laser Ultrasonic Propagation Imaging (UPI).
- Developed embedded firmware on ARM Cortex-M4 (LPC4370) for real-time signal acquisition and processing.
- Built amplifier, filter, and signal-generator circuits for ultrasonic actuation and sensing.
- Integrated Wi-Fi communication for wireless data transmission and monitoring.
- Created Qt-based desktop software for visualization and analysis of structural damage.

During B.E.

- GSM-based Automatic Meter Reading — built with an 8051 microcontroller, GSM modem, and C.
- FM Transmitter — built with a simple oscillator and Class-A amplifier.

REVIEWER & EDITORIAL EXPERIENCE

- Manuscript Reviewer — Smart Structures and Systems: An International Journal (Paper ID SSS07266J, 2025).
- Manuscript Reviewer — Advances in Engineering and Technology: An International Journal.
- Conference Reviewer — 1st IEEE International Conference on ICT and Photonics (ICTP 2026), Kathmandu, Nepal; Track: RF and Photonics.

AWARDS & HONORS

- ASABE Travel Scholarship (2025) — SDSU Department of Agricultural & Biosystems Engineering, to support participation in the 2025 ASABE Annual International Meeting, Toronto, Canada.
- NSF Travel Award (2023) — supported attendance at the USDA Multistate Meeting.

PROFESSIONAL TRAINING

- Structural Health Monitoring Using Statistical Pattern Recognition — Chonbuk National University (Aug 2014).
- Embedded Systems — Information Technology Society (Sep 2009).

REFERENCES

Prof. Lin Wei — South Dakota State University, USA | Lin.Wei@sdstate.edu

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